

TRACE-IDS: Zero-Shot Cross-Dataset Intrusion Detection Fails, and Few-Shot Labeling Recovers It

Rijan Rayamajhi
Independent Researcher
rayamajhirijan5051@gmail.com

Abstract—Machine-learning intrusion detectors reach near-perfect scores when trained and tested on one dataset, but deployment means running on a network the model never saw. We evaluate this transfer directly on two standardized NetFlow v3 datasets (NF-CICIDS2018 and NF-UNSW-NB15, 47 shared features). Zero-shot transfer fails completely: a Random Forest with 0.95 in-dataset F1 detects *zero* attacks on the other dataset (cross-dataset recall 0.00), and its 0.96 “accuracy” is an artifact of class imbalance. Per-dataset feature alignment, balanced class weights, and explanation-guided feature selection do not fix it. The practical fix is cheap supervision: adding only five labeled examples per class from the target network recovers 0.88 attack recall, ten give 0.93, and fifty give 0.99. Cross-dataset intrusion detection is not a representation problem to be solved once; it needs a small, ongoing amount of target labeling, and we quantify how little.

Index Terms—intrusion detection, cross-dataset generalization, few-shot learning, transfer learning, NetFlow

I. INTRODUCTION

Most machine-learning intrusion detectors are judged on one dataset: train on it, test on it, report F1 above 0.95. Deployment is not like that. A real detector meets traffic from a network it never trained on. We test that directly and find the benchmark numbers do not survive it: a Random Forest trained on NF-CICIDS2018 scores 0.95 F1 in-dataset, then catches *zero* attacks on NF-UNSW-NB15.

The usual repairs do not help. Rescaling each dataset to a common range, weighting the minority class, and selecting “transferable” features by SHAP all leave cross-dataset recall at zero. What does help is a little target data: a handful of labeled flows from the new network restores most of the detection. We measure how many are needed.

- On standardized NetFlow v3 data we show zero-shot cross-dataset detection fails outright (recall 0.00), and that accuracy hides this because the target is 96% benign.
- We show alignment, balanced weights, and invariant-feature selection do not recover it.
- We quantify a cheap fix: five labeled examples per class give 0.88 recall, ten give 0.93, fifty give 0.99 — so deployment on a new network needs very little labeling.

II. RELATED WORK

Cross-dataset generalization of ML-NIDS is known to be poor: within-dataset scores above 0.95 fall toward chance across datasets [1]. Domain-invariant representation methods

aim to close this gap [2], and SHAP has been used to study feature transfer [3]. Few-shot learning is the emerging practical direction, recovering strong cross-domain accuracy from a handful of target samples [4]. We contribute a clean, reproducible measurement on standardized NetFlow v3 data: the zero-shot failure, the fact that common fixes do not help, and a few-shot recovery curve.

III. METHOD

A. Datasets

We use NF-CICIDS2018-v3 and NF-UNSW-NB15-v3. Both share one standardized 55-column NetFlow schema, so no manual feature mapping is needed. We drop identifier columns (IP addresses, ports, timestamps, attack-category) that leak dataset identity, keeping 47 flow-statistic features and the binary label. NF-CICIDS2018 is subsampled to about 2M flows for tractability; NF-UNSW-NB15 is used in full (about 2.2M flows). The datasets are imbalanced: 13% attacks in CIC, 4% in UNSW.

B. Zero-Shot Transfer and the Fixes We Tried

A Random Forest (200 trees, fixed seed, balanced class weights) trains on 70% of CIC and is tested on the held-out 30% (same-dataset) and on all of UNSW (cross-dataset). We compare raw features against per-dataset alignment (z-scoring each dataset by its own statistics), and against explanation-guided invariant-feature selection (keep features important in both datasets with the same effect direction, by SHAP). Because the target is heavily imbalanced, we report recall and F1, not accuracy.

C. Few-Shot Recovery

We then add k labeled flows per class, drawn from a target pool, to the source training set and retrain, evaluating on a fixed held-out target test set. $k = 0$ is zero-shot. Both datasets are z-scored by their own statistics so they share a comparable space.

IV. RESULTS AND DISCUSSION

Table I shows the zero-shot picture. In-dataset F1 is 0.95, but cross-dataset recall is 0.00 for every configuration: the detector labels all UNSW traffic benign and catches no attacks. Its 0.96 cross-dataset “accuracy” is purely the benign fraction.

TABLE I
ZERO-SHOT TRANSFER, NF-CICIDS2018 $\rightarrow$ NF-UNSW-NB15. RECALL
IS ATTACK DETECTION.

Configuration	same F1	cross recall	cross F1	cros
No alignment (raw)	0.95	0.00	0.00	0.
Per-dataset alignment	0.95	0.00	0.00	0.
Alignment + invariant selection	0.93	0.00	0.00	0.

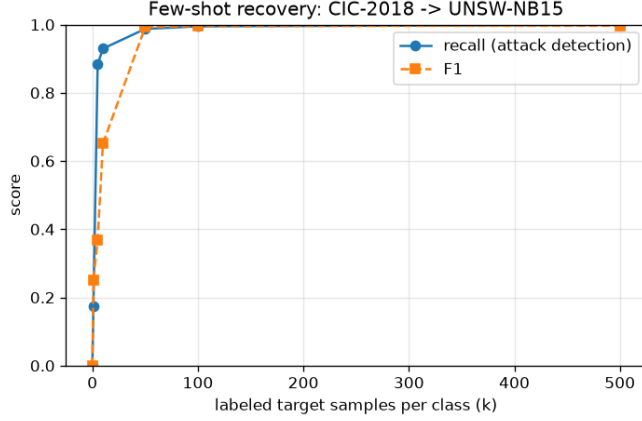


Fig. 1. Few-shot recovery. Zero-shot recall is 0.00; a handful of labeled target flows per class restores detection (five $\rightarrow$ 0.88, ten $\rightarrow$ 0.93, fifty $\rightarrow$ 0.99).

Alignment changes nothing, balanced weights do not rescue recall, and invariant-feature selection only lowers accuracy without raising recall. On standardized features, naive transfer simply does not work.

Figure 1 shows the fix. Adding just five labeled examples per class from the target network raises attack recall from 0.00 to 0.88; ten give 0.93, fifty give 0.99. Detection recovers almost completely with a trivial amount of target labeling.

The takeaway is practical. Cross-dataset intrusion detection is not a representation that can be learned once and reused; the attack signatures differ too much between networks (Fig. 2). But it does not need full relabeling either — a few labeled flows per class on the new network are enough to redeploy a working detector.

V. CONCLUSION

On standardized NetFlow v3 data, zero-shot cross-dataset intrusion detection fails completely, and feature alignment, balanced weights, and invariant-feature selection do not fix it. Few-shot labeling does: five to fifty labeled examples per class recover 0.88 to 0.99 attack recall. The study is limited to two datasets and a Random Forest; offline, no real-time or adversarial testing. Future work: extend to a leave-one-dataset-out benchmark across more NetFlow datasets, and test few-shot with deeper models.

REFERENCES

- [1] M. Cantone, C. Marrocco, and A. Bria, “On the Cross-Dataset Generalization of Machine Learning for Network Intrusion Detection,” *IEEE Access*, vol. 12, 2024. arXiv:2402.10974.

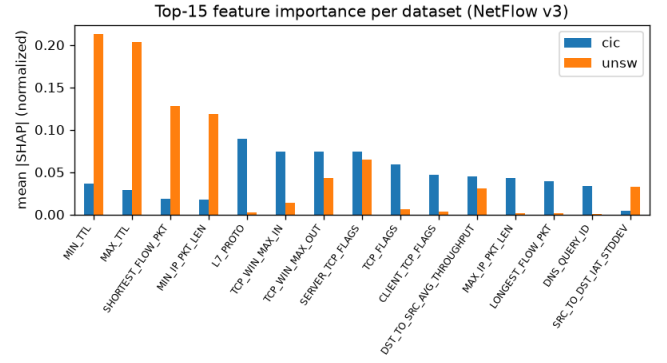


Fig. 2. Per-dataset SHAP feature importance. The features the model leans on differ between datasets, which is why a source-only model does not transfer.

- [2] S. Layeghy, M. Baktashmotlagh, and M. Portmann, “DI-NIDS: Domain Invariant Network Intrusion Detection System,” 2022. arXiv:2210.08252.
- [3] C. Kılıç and G. Şengül, “SHAP-Guided Feature Selection for Cross-Dataset Generalization in Network Intrusion Detection Systems,” Atılım University, 2026.
- [4] “Few-shot network intrusion detection method based on multi-domain fusion and cross-attention,” *PLOS ONE*, 2025. [verify authors before submitting]